Analogue I-V Curve Tracer Module

1 Features

- Standalone hardware topology (No MCU required)
- Synchronized 5-step staircase base current generator utilizing CD4017B logic
- Integrated triangle wave collector sweep
- 4-layer, low-noise PCB design (60mm $\times$ 60mm)
- Test Points accounted for on important nodes
- Battery holder for 9V alkaline battery
- Reverse polarity and ESD protection

2 Applications

- Discrete semiconductor characterization (NPN, NMOS, Diodes)
- Component matching and tolerance verification
- Educational laboratory demonstrations
- Reactive component (Capacitor/Inductor) loop analysis

3 General Description

The Analog I-V Curve Tracer is a compact evaluation module designed to generate standard characteristic curves for two- and three-terminal devices on any oscilloscope equipped with an X-Y mode. By utilizing a continuous triangle wave generated by an op-amp integrator and a decade counter used to generate a staircase wave, the module safely steps base current while sweeping collector voltage.

Note: *Board can likely be made into 2-layers with ground plane on bottom if you would like, to save cost*

The 4-layer layout (SIG $\rightarrow$ GND $\rightarrow$ GND $\rightarrow$ SIG) aids signal integrity for microamp-level current steps. It is built to be powered by a 9V alkaline battery such as these from Amazon.

There is a GitHub repository for this project, containing all files for the project.

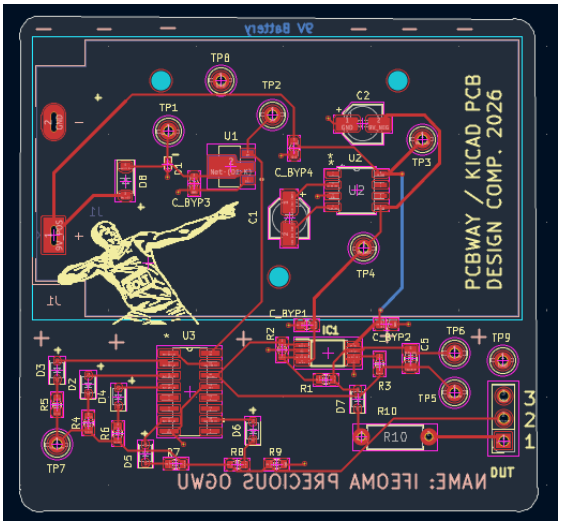

Figure 0. Routing of the PCB

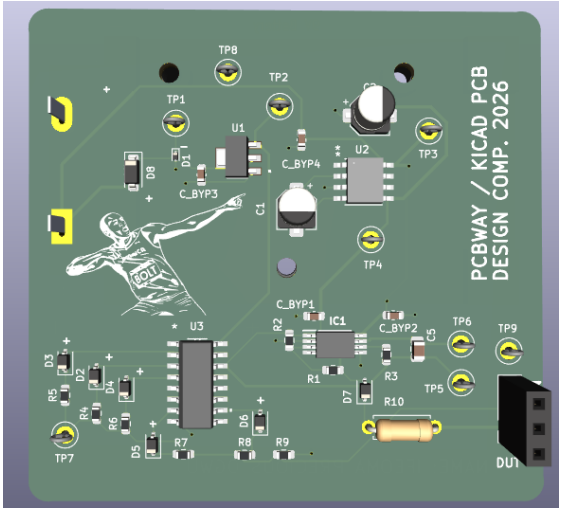

Figure 1. Top view of the 60x60mm assembled module.

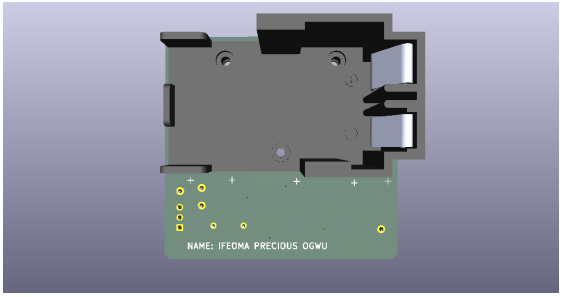

Figure 2. Bottom view of the 60x60mm assembled module.

4 Specifications

4.1 Absolute Maximum Ratings

Stresses beyond those listed may cause permanent damage to the device.

Parameter	Conditions	Max
Supply Voltage	Continuous	9V
Module Power Dissipation	$T_A = 25°$C	1.1 mW
DUT Power Dissipation	$T_A = 25°$C	0.2 W

4.2 Electrical Characteristics

$V_{CC} = +9$*V*, $V_{EE} = -9$*V*, $V_{DD} = 5$*V*, $T_A = 25°$*C unless otherwise noted.*

Parameter	Min	Typ	Max
Power Supply			
Operating Voltage	7V	9V	10V
Source Resistance	–	55	–
Quiescent Current	–	2.4 mA	–
Sweep Generator			
Sweep Freq.	–	100Hz	–
Sweep Output (V_{pk-pk})	–	9V	–
Square Wave (V_{pk-pk})	–	16V	–
Base Drive			
Number of Steps	–	5	–
Resistance	450	–	–
Capacitance	10F	–	1F
Inductance	47mH	–	–

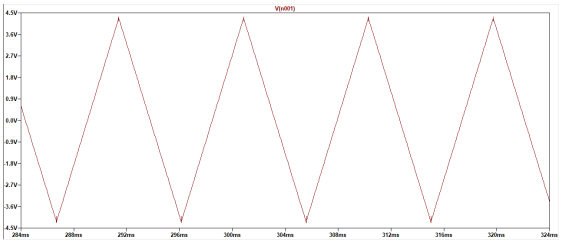

Figure 5. Triangle Wave From Simulation.

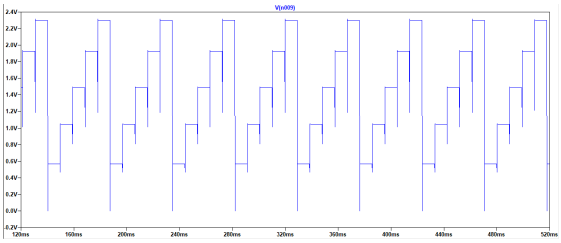

Figure 6. Staircase Wave From Simulation.

5 Detailed Description and Simulations

Simulations were conducted in LTSpice to validate design. A comparator op-amp circuit generates a 100Hz square wave that is fed into a comparator circuit generating a triangle wave. The op-amps used are the TL072H.

The square wave is fed into the clock signal of the CD4017B and the decade counter is used to generate the staircase wave. The staircase wave output is fed into a npn or NMOS (if that is the current DUT) through an external base / gate resistor.

The triangle wave is fed into a diode and resistor in series to pass only positive values of the triangle wave. This resistor connection is connected to the collector / drain of a npn or NMOS transistor of the top of a resistor, capacitor, inductor or positive side of a diode (schottky, zener etc) depending on the DUT.

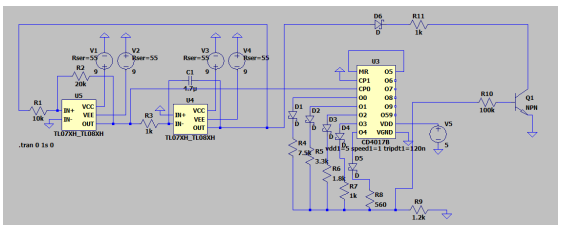

Figure 3. LTSpice Simulation.

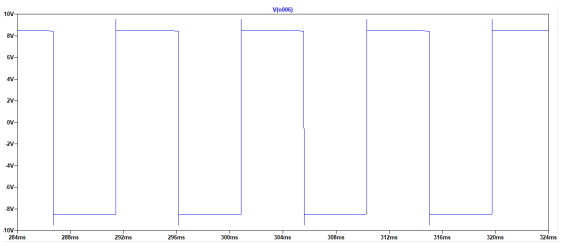

Figure 4. Square Wave From Simulation.

For the PCB:

The 9V battery is connected through a 1N5817 diode, used for reverse polarity protection, then a 11V TVS diode is connected from the cathode of 1N5817 to ground for ESD protection.

The line continues then to connect to the 5V LDO HT7350-A used to connect to VDD of the CD4017B. The 9V battery is also connected to an ICL7660 charge pump that gives us -9V which we can use ±9V for the op-amp. From there, the connections are basically same as in LTSpice. There are 0.1 bypass capacitors added to the input pins of the ICs to reduce / prevent noise.

The outputs of the entire circuit needed for the DUT is connected to a 3-pin 2.54mm header socket. There are also test points placed, that can be used for debugging or other purposes.

Note: *There's errors in DRC due to the components being on top of the battery holder. However, looking sideways, you can solder and cut the excess off enough to not interfere*

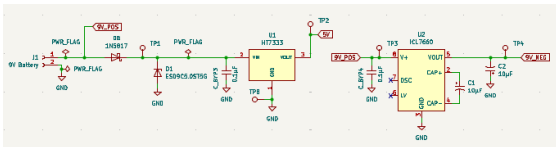

Figure 7. Positve and negative 9V generation, as well as 5V.

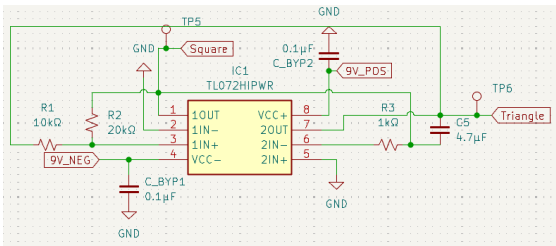

Figure 8. Square and triangle wave generation from op-amps.

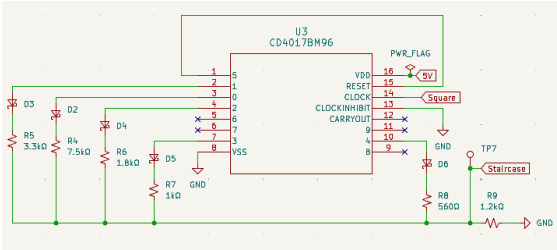

Figure 10. Staircase wave generation from decade counter.

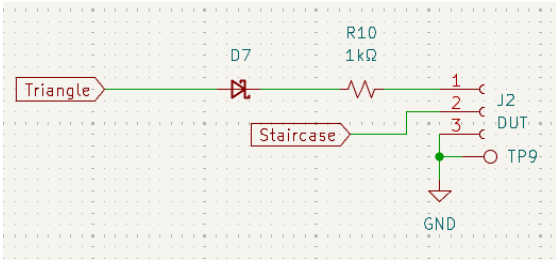

Figure 11. Outputs for DUT.

6 Naming

For use of this module, these are the relevant pins and their uses:

- **TP1:** 9V (positive). Placed in a such a way, it can be used to verify if the D8 is working properly and in right direction.
- **TP2:** 5V
- **TP3:** 9V (positive)
- **TP4:** 9V (negative)
- **TP5:** Square wave
- **TP6:** Triangle wave
- **TP7:** Staircase wave
- **TP8:** GND
- **TP9:** GND. This test point is located next to the DUT socket so you can use to ground your oscilloscope probe.

  For the 3-pin socket that you can place your DUT into:

- **1:** Connected to the series diode and 1k resistor connected from the triangle wave to the "top" of the DUT
- **2:** Staircase wave output. Use for NMOS and npn.
- **3:** GND

7 Hardware Interface

How to connect each DUT:

7.1 Resistor

- Connect one end of the resistor to R10 and the other end to GND.

7.2 Capacitor

- Connect one end of the capacitor (if polarized, connect the positive) to R10 and the other end to GND

7.3 Inductor

- Connect one end of the inductor to R10 and the other end to GND.

7.4 Diode

- Connect the anode (positive) of the diode to R10 and the cathode (negative) to GND.

7.5 npn

- Connect the collector to R10, connect the base to an external resistor which you then connect to pin 2 of the DUT socket, connect the emitter to GND.

7.6 NMOS

- Connect the drain to R10, connect the gate to an external resistor which you then connect to pin 2 of the DUT socket, connect the source to GND.

How to connect DUT to oscilloscope:

If you have a probe that can measure current such as this *(any products links in this document are not recommendations or endorsements [unless they are the ICs and components used in the PCB], they are simply examples)* from Amazon then:

- Connect one probe (this will be X-axis) to the part of your DUT that is connected to pin 1.
- Connect a clamp probe (this will be Y-axis) around the wire or pin that connects your Device Under Test (DUT) to pin 1
- Use XY mode to plot current vs. voltage
- **GND:** Ensure oscilloscope ground clips are tied strictly to the board's designated common ground to prevent shorting.

If you do not have a current clamp probe:

- Connect a small shunt resistor (eg 0.1 or 1) in series with pin 1 to your DUT. You maybe just use the on-board R10 as your shunt. This has not been tested yet, however.
- Connect a probe (this will be Y-axis) to one side of the resistor and the ground clip to the other. *(Note: Ensure the ground clip is connected to the side closest to ground to avoid shorting your circuit, or use a differential measurement.)*
- On the oscilloscope, use the Math function to divide the channel voltage by the shunt resistance value.
- Connect another probe (this will be X-axis) to the part of your DUT that is connected to pin 1.
- Use XY mode to plot current vs. voltage
- **GND:** Ensure oscilloscope ground clips are tied strictly to the board's designated common ground to prevent shorting.

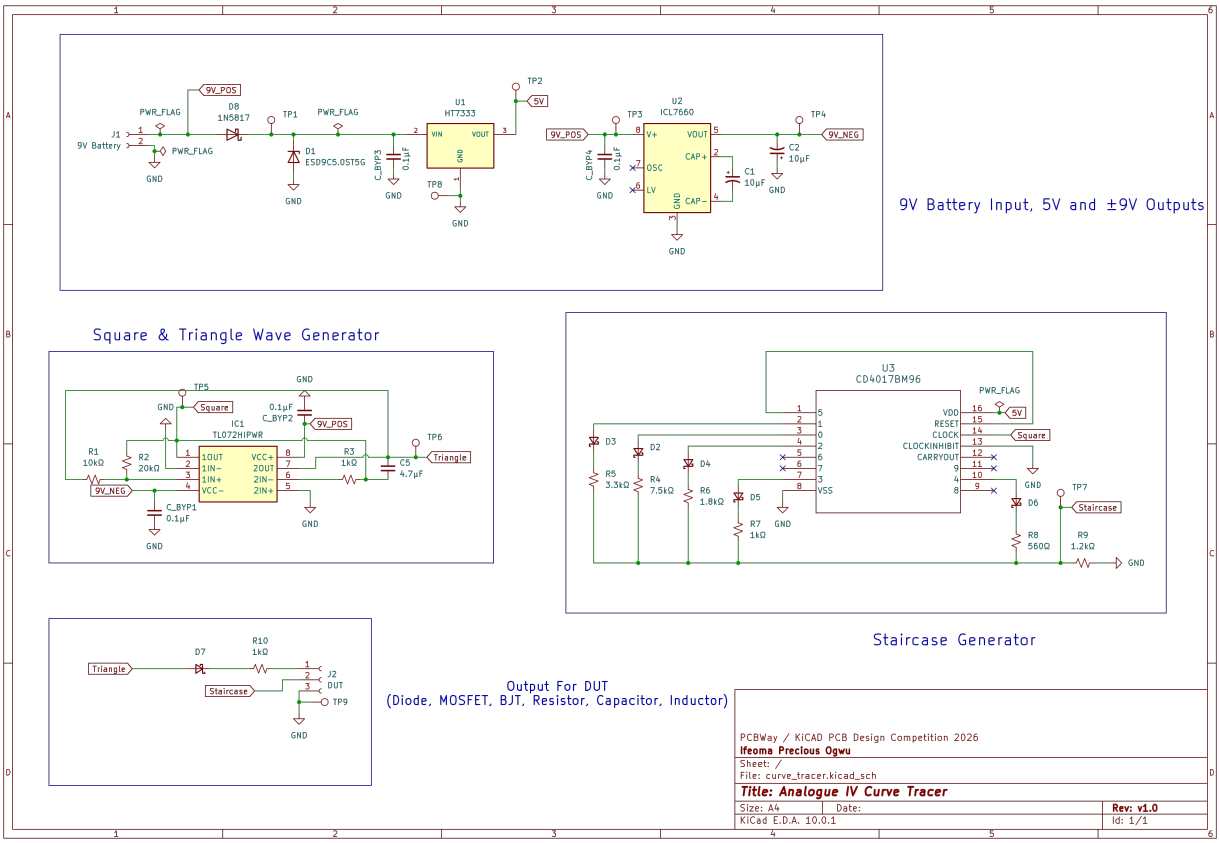

Figure 1: Figure 12. Full KiCAD Schematic

Typical Performance Characteristics

The following waveforms represent typical operation using standard off-the-shelf components.

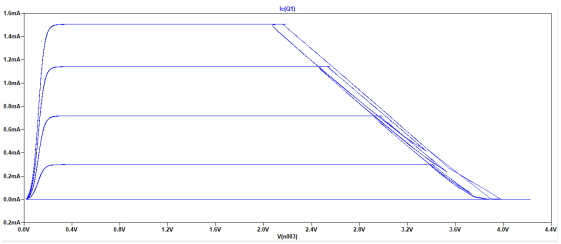

Figure 13. NPN Transistor Characteristic Sweep (I_c vs V_{ce}).

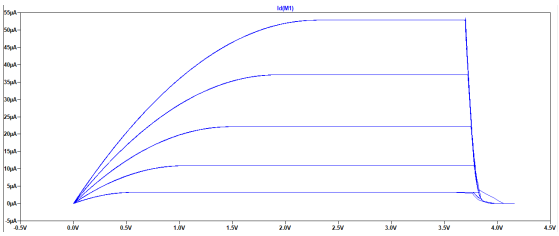

Figure 13. NMOS Transistor Characteristic Sweep (I_G vs V_{DS}).

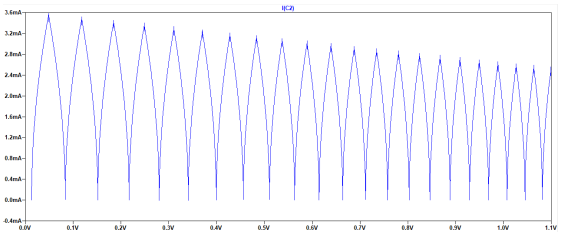

Figure 15. Reactive Load Analysis: 100μF Capacitor showing expected curve

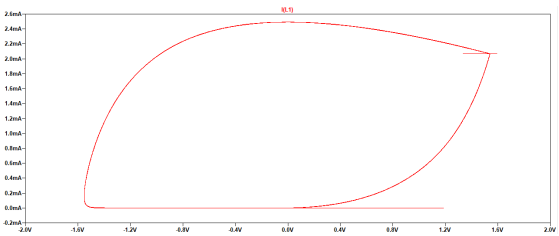

Figure 16. Reactive Load Analysis: 1H Inductor showing standard elliptical response.

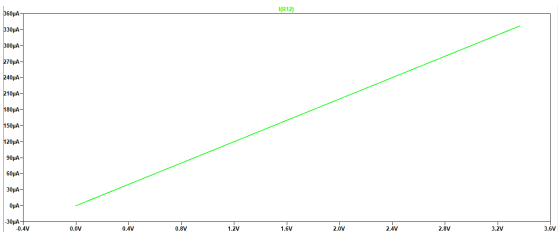

Figure 17. $10k$ resistor showing expected straight-line graph.

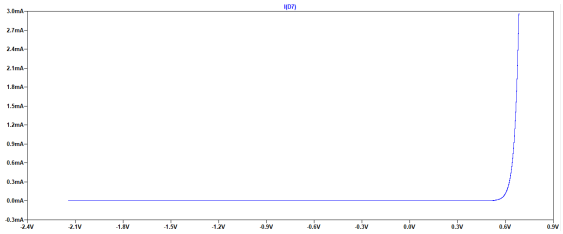

Figure 18. Diode showing expected curve.

8 BOM

The components were selected for a balance between high functionality and lower costs. The current BOM costs a total of $11.88 USD = 8.82 = 10.16 = $16.24 CAD = $207.24 MXN. The components were mainly sourced from DigiKey. There are also components sourced from LCSC and Amazon. You can source components from wherever you please, as long as it is the same footprint and make for the main ICs and as long as you are fine with your selected tolerances for capacitors and resistors.

Note: *All resistors are 0.5W rated. You can get resistors rated for more power, but it is not recommended to get resistors rated for lower power.*

The BOM is posted on the GitHub page for this project.

9 Design Limitations

While optimized for standard NPN and NMOS topologies, this hardware revision utilizes a unidirectional positive voltage sweep. Consequently, PNP and PMOS characterization will result in inverted/mirrored plots and may not fully forward-bias internal junctions correctly.

You are welcome to attempt to rectify this issue.

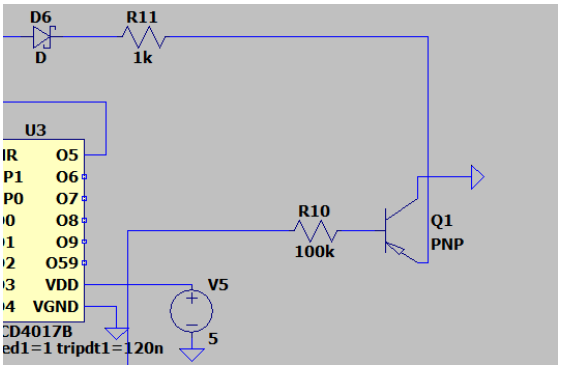

Figure 19. LTSpice Setup For pnp Or PMOS

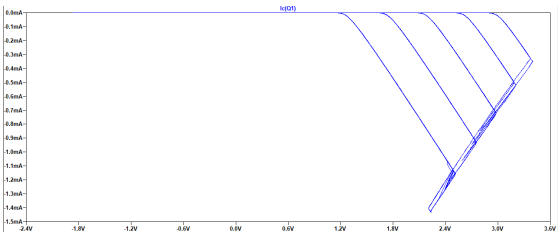

Figure 20. pnp curve.

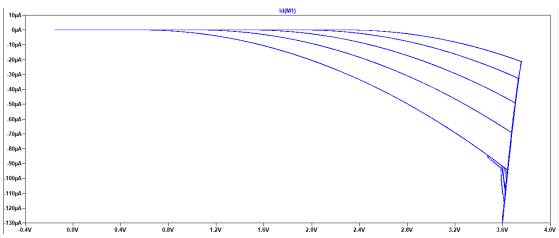

Figure 21. PMOS curve.

Easter Egg: Why did I choose Usain Bolt specifically on my silkscreen?

10 Concluding Notes

Here are some of the resources I used to learn more about how to design and build a curve tracer:

- Building A Curve Tracer
- Building A Curve Tracer v3
- Design & Build a Stair-Step Waveform Generator
- Analog Curve Tracer - MIT Report
- Op-Amp Comparator Circuit
- Op-Amp Integrator Circuit

Most of the websites are in pdf format on the GitHub page for this project.

Using the online PCBWay instant quote:

For a PCBA, each board would cost about $29 Without PCBA, each board would cost about $17